Mathematical Methods for Finance

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Module 5 - Lesson 16

Stochastic integration

From simple predictable strategies to the Itô integral

Learning outcomes for Lesson 16

By the end of the lesson students should be able to

1. define the Itô integral first for simple predictable processes;
2. compute expectations and variances of stochastic integrals;
3. use Itô isometry correctly;
4. understand the extension from simple processes to L^2 integrands;
5. recognise local L^2 ideas and integration with respect to continuous semimartingales.

Intuition: Operating rule

Before computing, identify the object: pathwise variation, ordinary integral, quadratic variation, stochastic integral, Itô process or stochastic differential rule.

How today is organised

Block 1

14:30–15:20
Why ordinary sums are replaced by adapted left-point sums.

Block 2

15:35–16:25
Simple processes, Itô isometry and construction in L^2.

Block 3

16:40–17:30
Localisation, semimartingale integrators and exam-style computations.

Exercise: During the lesson

Every exercise should end with one sentence explaining what the formula means.

Block 1

What the Itô integral is trying to do

Starting point

Goal

We want to give meaning to

$$\int_0^T H_t \, dB_t,$$

where B is Brownian motion and H is a process observed over time.

Intuition: Key rule

In Itô integration, the integrand must use information available before the Brownian increment is realised.

Why left endpoints?

Definition: Itô-type sum

For a partition π, the Itô sum has the form

$$\sum_i H_{t_{i-1}}\left(B_{t_i} - B_{t_{i-1}}\right).$$

Intuition: No anticipation

The coefficient $H_{t_{i-1}}$ is known before the increment $B_{t_i} - B_{t_{i-1}}$ occurs.

Counterexample: Do not use future information

Using H_{t_i} instead of $H_{t_{i-1}}$ generally changes the limit.

Adapted and predictable

Definition: Adapted process

A process H_t is adapted to $(\mathcal{F}_t)$ if H_t is $\mathcal{F}_t$-measurable for every t.

Definition: Predictable step process

A simple predictable process has the form

$$H_t = \sum_{k=0}^{n-1} H_k \mathbf{1}_{(t_k, t_{k+1}]}(t),$$

where H_k is $\mathcal{F}_{t_k}$-measurable.

First construction: simple processes

Definition: Itô integral for simple predictable processes

If

$$H_t = \sum_{k=0}^{n-1} H_k \mathbf{1}_{(t_k, t_{k+1}]}(t),$$

then

$$\int_0^T H_t \, dB_t = \sum_{k=0}^{n-1} H_k \big(B_{t_{k+1}} - B_{t_k}\big).$$

This is the foundation

All later definitions are obtained by approximation from this simple case.

Worked example: deterministic step integrand

Exercise: ⋆⋆

Let

$$H_t = 2\mathbf{1}_{(0,1]}(t) - \mathbf{1}_{(1,2]}(t).$$

Compute $\int_0^2 H_t \, dB_t$.

Worked example: Solution

By the definition for simple processes,

$$\int_0^2 H_t \, dB_t = 2(B_1 - B_0) - 1(B_2 - B_1) = 2B_1 - B_2 + B_1 = 3B_1 - B_2.$$

Since $B_0 = 0$, the expression is $3B_1 - B_2$.

Expectation of the simple integral

Theorem: Mean zero

If H is square-integrable and predictable, then

$$\mathbb{E}\left[\int_0^T H_t \, dB_t\right] = 0.$$

Proof idea: Simple case

For a simple process,

$$\mathbb{E}\big[H_k(B_{t_{k+1}} - B_{t_k})\big] = 0,$$

because H_k is known at time t_k and the Brownian increment has conditional mean zero.

Itô isometry

Theorem: Itô isometry

For square-integrable predictable H,

$$\mathbb{E}\left[\left(\int_0^T H_t\, dB_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2\, dt\right].$$

Intuition: Meaning

The variance of the stochastic integral is controlled by the accumulated squared size of the integrand.

Exercise: variance of a deterministic integrand

Exercise: ⋆⋆

Let $H_t = t$ on $[0, T]$. Compute

$$\mathbb{E}\left[\int_0^T t\, dB_t\right] \quad \text{and} \quad \operatorname{Var}\left(\int_0^T t\, dB_t\right).$$

Worked example: Solution

The mean is zero. By Itô isometry,

$$\operatorname{Var}\left(\int_0^T t\, dB_t\right) = \int_0^T t^2\, dt = \frac{T^3}{3}.$$

Distribution with deterministic integrand

Theorem: Gaussian case

If h is deterministic and square-integrable, then

$$\int_0^T h(t)\, dB_t \sim N\left(0, \int_0^T h(t)^2\, dt\right).$$

Exercise: $\star\star$

Find the distribution of $\int_0^1 2t\, dB_t$.

Worked example: Solution

The variance is $\int_0^1 4t^2\, dt = 4/3$. Hence

$$\int_0^1 2t\, dB_t \sim N(0, 4/3).$$

Block 2

Construction in L^2 and core computations

The space of admissible integrands

Definition: L^2 integrability

An adapted process H is square-integrable on $[0, T]$ if

$$\mathbb{E}\left[\int_0^T H_t^2 \, dt\right] < \infty.$$

Construction idea

Approximate H by simple predictable processes $H^{(n)}$ and define

$$\int_0^T H_t \, dB_t = L^2\text{-limit of } \int_0^T H_t^{(n)} \, dB_t.$$

Why the construction works

Proof idea: Cauchy mechanism

If $H^{(n)} \to H$ in $L^2(\Omega \times [0,T])$, then Itô isometry gives

$$\mathbb{E}\left[\left(\int_0^T (H_t^{(n)} - H_t^{(m)})\, dB_t\right)^2\right] = \mathbb{E}\left[\int_0^T (H_t^{(n)} - H_t^{(m)})^2\, dt\right].$$

Thus the integrals are Cauchy in $L^2(\Omega)$.

Intuition: Connection with Module 1

Completeness is what guarantees the existence of the limit of approximating integrals.

Stochastic integral as a martingale

Theorem: Integral process

If

$$M_t = \int_0^t H_s \, dB_s,$$

then, under square-integrability assumptions, (M_t) is a martingale.

Quadratic variation

Its quadratic variation is

$$\langle M \rangle_t = \int_0^t H_s^2 \, ds.$$

Exercise: martingale and quadratic variation

Exercise: $\star\star\star$

Let

$$M_t = \int_0^t (1+s)\, dB_s.$$

Compute $\mathbb{E}[M_t]$, $\mathrm{Var}(M_t)$ and $\langle M \rangle_t$.

Worked example: Solution

The mean is zero. By Itô isometry,

$$\mathrm{Var}(M_t) = \int_0^t (1+s)^2\, ds = t + t^2 + \frac{t^3}{3}.$$

Also

$$\langle M \rangle_t = \int_0^t (1+s)^2\, ds.$$

The integral $\int_0^t B_s\, dB_s$

Example: A central example

The stochastic integral

$$\int_0^t B_s\, dB_s$$

is well-defined because B_s is adapted and square-integrable on finite intervals.

Preview

Using Itô formula in Lesson 17, we will obtain

$$\int_0^t B_s\, dB_s = \frac{1}{2}(B_t^2 - t).$$

Exercise: second moment of $\int B\,dB$

Exercise: $\star\star\star$

Use Itô isometry to compute

$$\mathbb{E}\left[\left(\int_0^T B_t\,dB_t\right)^2\right].$$

Worked example: Solution

By Itô isometry,

$$\mathbb{E}\left[\left(\int_0^T B_t\,dB_t\right)^2\right] = \mathbb{E}\left[\int_0^T B_t^2\,dt\right] = \int_0^T \mathbb{E}[B_t^2]\,dt = \int_0^T t\,dt = \frac{T^2}{2}.$$

Ordinary time integral versus Itô integral

Ordinary integral

$$\int_0^T H_t \, dt$$

accumulates the value of H_t over time.

Itô integral

$$\int_0^T H_t \, dB_t$$

accumulates H_t against random Brownian increments.

Exercise: Quick check $\star$

For $H_t = 1$, compute both $\int_0^T 1 \, dt$ and $\int_0^T 1 \, dB_t$.

Worked example: Answer

They are T and B_T, respectively.

A warning: Itô integral is not pathwise Riemann–Stieltjes

Counterexample: False idea

Do not define $\int H\,dB$ path by path using bounded variation. Brownian paths do not have bounded variation.

Intuition: Correct idea

The Itô integral is defined by probabilistic limits of adapted sums. The time at which H is sampled matters.

Simple random integrand

Exercise: $\star\star\star$

Let $H_t = B_1 \mathbf{1}_{(1,2]}(t)$. Compute

$$\int_0^2 H_t \, dB_t.$$

Is H predictable on $(1, 2]$?

Worked example: Solution

Since B_1 is known at time 1,

$$\int_0^2 H_t \, dB_t = B_1(B_2 - B_1).$$

The integrand is predictable on $(1, 2]$ because the coefficient is $\mathcal{F}_1$-measurable.

Independence check

Exercise: $\star\star\star$

Using the previous exercise, compute

$$\mathbb{E}\big[B_1(B_2 - B_1)\big].$$

Worked example: Solution

B_1 is independent of $B_2 - B_1$, and $\mathbb{E}[B_2 - B_1] = 0$. Therefore

$$\mathbb{E}\big[B_1(B_2 - B_1)\big] = \mathbb{E}[B_1]\mathbb{E}[B_2 - B_1] = 0.$$

Block 3

Localisation and semimartingale integration

Why local L^2 is needed

Motivation

Some integrands are not globally square-integrable, but become square-integrable after stopping before they become too large.

Definition: Local square integrability, informal

A process H is locally square-integrable if there exist stopping times $\tau_n \uparrow \infty$ such that

$$\mathbb{E}\left[\int_0^{T\wedge\tau_n} H_t^2\, dt\right] < \infty$$

for every n.

Stopping before trouble

Intuition: Localisation

Instead of requiring a process to be controlled forever, we stop it at levels where it is controlled, do the calculation, and then let the stopping level increase.

Example: Typical stopping time

$$\tau_n = \inf\{t \geq 0 : |H_t| \geq n\}.$$

Before τ_n, the process is bounded by n.

Integration with respect to continuous martingales

Theorem: General isometry idea

If M is a continuous square-integrable martingale, stochastic integration can be built with respect to M and

$$\mathbb{E}\left[\left(\int_0^T H_t \, dM_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2 \, d\langle M\rangle_t\right].$$

Brownian case

For $M = B$, $d\langle B\rangle_t = dt$.

Integration with respect to semimartingales

Definition: Semimartingale integral, informal

If $X = M + A$ is a semimartingale, then

$$\int H\,dX = \int H\,dM + \int H\,dA.$$

The first term is stochastic; the second is finite-variation pathwise integration.

Intuition: Two pieces

The martingale part requires Itô integration. The finite-variation part is classical.

Exercise: integrate against drift plus Brownian motion

Exercise: $\star\star\star$

Let $X_t = t + B_t$. Write

$$\int_0^T H_t \, dX_t$$

in terms of ordinary and Itô integrals.

Worked example: Solution

Since $X_t = A_t + M_t$ with $A_t = t$ and $M_t = B_t$,

$$\int_0^T H_t \, dX_t = \int_0^T H_t \, dt + \int_0^T H_t \, dB_t.$$

A complete computation

Exercise: $\star\star\star$

Let $X_t = 2t + 3B_t$ and $H_t = t$. Compute the mean and variance of the martingale part of

$$\int_0^T H_t \, dX_t.$$

Worked example: Solution

We have

$$\int_0^T t \, dX_t = \int_0^T 2t \, dt + \int_0^T 3t \, dB_t.$$

The martingale part is $\int_0^T 3t \, dB_t$, with mean zero and variance

$$\int_0^T 9t^2 \, dt = 3T^3.$$

Exam exercise: simple stochastic integral

Exercise: $\star\star\star$

Let

$$H_t = 1_{(0,1]}(t) + 2B_1 1_{(1,2]}(t).$$

Compute $\int_0^2 H_t \, dB_t$ and its expectation.

Worked example: Solution

The integral is

$$(B_1 - B_0) + 2B_1(B_2 - B_1) = B_1 + 2B_1(B_2 - B_1).$$

The expectation is zero: $\mathbb{E}[B_1] = 0$ and B_1 is independent of $B_2 - B_1$ with mean zero.

Exam exercise: Itô isometry with random integrand

Exercise: ⋆⋆⋆

Compute

$$\mathbb{E}\left[\left(\int_0^T B_t^2\, dB_t\right)^2\right].$$

Use the Gaussian moment $\mathbb{E}[B_t^4] = 3t^2$.

Worked example: Solution

By Itô isometry,

$$\mathbb{E}\left[\left(\int_0^T B_t^2\, dB_t\right)^2\right] = \mathbb{E}\left[\int_0^T B_t^4\, dt\right] = \int_0^T 3t^2\, dt = T^3.$$

Exercise: conditional mean of an Itô integral

Exercise: $\star\star\star$

Let $H_t = H_0$ on $(0, 1]$, where H_0 is $\mathcal{F}_0$-measurable and square-integrable. Compute

$$\mathbb{E}\left[\int_0^1 H_t \, dB_t \mid \mathcal{F}_0\right].$$

Worked example: Solution

The integral is $H_0(B_1 - B_0) = H_0 B_1$. Since the Brownian increment has conditional mean zero,

$$\mathbb{E}[H_0 B_1 \mid \mathcal{F}_0] = H_0 \mathbb{E}[B_1 \mid \mathcal{F}_0] = 0.$$

Exercise: non-anticipation check

Exercise: ⋆⋆

Is $H_t = B_2 \mathbf{1}_{(1,2]}(t)$ a valid simple predictable integrand on $(1, 2]$?

Worked example: Solution

No. On $(1, 2]$, the coefficient must be known at time 1. But B_2 is not $\mathcal{F}_1$-measurable. It uses future information.

Exercise: stochastic integral with two intervals

Exercise: $\star\star\star$

Let

$$H_t = 2\mathbf{1}_{(0,1]}(t) + B_1\mathbf{1}_{(1,3]}(t).$$

Compute $\int_0^3 H_t\, dB_t$.

Worked example: Solution

By the simple-process definition,

$$\int_0^3 H_t\, dB_t = 2(B_1 - B_0) + B_1(B_3 - B_1) = 2B_1 + B_1(B_3 - B_1).$$

Exercise: second moment with a two-step integrand

Exercise: $\star\star\star$

For

$$H_t = 2\mathbf{1}_{(0,1]}(t) + \mathbf{1}_{(1,3]}(t),$$

compute $\mathbb{E}[(\int_0^3 H_t\, dB_t)^2]$.

Worked example: Solution

By Itô isometry,

$$\mathbb{E}\left[\left(\int_0^3 H_t\, dB_t\right)^2\right] = \int_0^3 H_t^2\, dt = \int_0^1 4\, dt + \int_1^3 1\, dt = 6.$$

Exercise: from integral to normal law

Exercise: ⋆⋆

Find the law of $I = \int_0^2 (3 - t)\, dB_t$.

Worked example: Solution

The integrand is deterministic, so I is Gaussian with mean zero and variance

$$\int_0^2 (3 - t)^2\, dt = \int_0^2 (9 - 6t + t^2)\, dt = 18 - 12 + \frac{8}{3} = \frac{26}{3}.$$

Thus $I \sim N(0, 26/3)$.

Mini-workshop: choose the right method

Exercise: ⋆⋆

For each task, choose the tool:

1. compute $\int_0^2 H_t \, dB_t$ for a step process;
2. compute $\mathbb{E}[(\int_0^T H_t \, dB_t)^2]$;
3. decide whether $H_t = B_3$ is allowed on $(1, 2]$;
4. split $\int H \, d(t + B_t)$ into two pieces.

Worked example: Answer

Simple-process definition; Itô isometry; predictability check; semimartingale decomposition.

Common mistakes

Counterexample: Mistakes to avoid

- Treating $\int H\,dB$ as an ordinary Riemann integral.
- Using right endpoints without noticing that the integral changes.
- Forgetting adaptedness or predictability.
- Computing $\mathbb{E}[\int H\,dB]$ by integrating $\mathbb{E}[H]$ against dB.
- Applying Itô isometry without squaring the integrand.

End-of-lesson checklist

Can you do these?

1. Define the Itô integral for simple predictable integrands.
2. Compute stochastic integrals of step processes.
3. Use Itô isometry to compute second moments.
4. Identify ordinary and stochastic pieces in semimartingale integrals.
5. Explain why no anticipation is essential.

Exercise: Suggested practice

Use the Module 5 exercise bank: simple Itô integrals, Itô isometry, local L^2 and semimartingale integration.